Somayeh Ghaneameleh



MSc Environmental Engineering | Prospective PhD Researcher

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Research Profile

MSc graduate in Biodiversity Management & Conservation with strong expertise in molecular phylogenetics, phylogeography, climate change, population genetics, and conservation genetics. Experienced in full field-to-lab molecular workflows including DNA extraction, PCR, gel electrophoresis, sequencing preparation, and multi-gene mitochondrial analyses. Skilled in Bayesian and Maximum Likelihood phylogenetic inference (MrModeltest, BEAST, MrBayes, and RAxML), population genetic analyses (DnaSP, Arlequin, Haplotype Network, and TCS), and geographic interpretation using GIS tools. Highly motivated to pursue research in evolutionary biology, molecular ecology, and conservation, with a strong interest in developing new molecular and genomic approaches.

Research Interests

- **Molecular phylogenetics and evolutionary history**
- **Phylogeography and demographic history (glacial cycles, refugia, geological events, and geographic structure)**
- **Population genetics and conservation units**
- **Conservation genetics and conservation priorities**
- **Climate change and ecological niche modelling as sources of evidence**

Education

2017-2020. M.Sc. in Science and Environmental Engineering- Biodiversity Management and Conservation

- **Shahid Beheshti University, Tehran, Iran (www.sbu.ac.ir)** **GPA: (17.60/20)**

  Thesis: Phylogeography of Three-lined Lizard (*Lacerta media*) in Iran

  Thesis Grade: Excellent (equivalent to 19/20)

  Supervisor: Dr. Faraham Ahmadzadeh (f_ahmadzade@sbu.ac.ir)

2010-2014. B.Sc. in Natural Resources Engineering- Environment

Kurdistan University, Kurdistan, Iran (www.uok.ac.ir) **GPA: (15.16/20)**

Project: Explaining the vital components, geography and biological threats of *Neurergus derjugini*

Supervisor: Dr. Hanyeh Ghaffari (h.ghaffari@uok.ac.ir)

Honors

Ranked 25th among >2500 participants in the Iranian National University Entrance Exam, Science and Environmental Engineering **(MSc) 2017.**

Publication

- **Ghane-Ameleh, S.**, Khosravi, M., Saberi-Pirooz, R., Ebrahimi, E., Asadi Aghbolaghi, M., & Ahmadzadeh, F. **(2021)**. "Mid-Pleistocene Transition as a Trigger for Diversification in the Irano-Anatolian Region: Evidence Revealed by Phylogeography and Distribution Pattern of the Eastern Three-Lined Lizard. **Global Ecology and Conservation**. https://doi.org/10.1016/j.gecco.2021.e01839
- Ahmadi, M., Hemami, M.-R., Kaboli, M., **Ghane-Ameleh, S.**, & Malekian, M. **(2025)**. "Conservation Biogeography of Mountain Vipers: A Phylogenetic Niche Modelling Approach." **Diversity and Distributions**. https://doi.org/10.1111/ddi.13955

Research Experience

Molecular Ecology Laboratory – Shahid Beheshti University

Research experience: 2 years

- Conducted molecular research under the supervision of Dr. Faraham Ahmadzadeh as part of a research team.
- Performed DNA extraction and PCR using laboratory equipment including Thermomixer, Microcentrifuge, Thermocycler (Bio-Rad), electrophoresis (Safe-Red™), and gel documentation systems.
- Worked collaboratively with other researchers during laboratory and research activities.

Skills

- **Field Working and Laboratory Skills**

Sampling with team working in the field, muscle tissue preservation in ethanol for long-term storage, DNA extraction, PCR, Nanodrop Spectrometry for DNA qualification, Electrophoresis (Agarose gel with Safe-Red) for suitable products, preparing sequences for sequencing and freezing genetic samples.

- **Software**

Geneious Prime, Mafft, MrModeltest, SequenceMatrix, MrBayes, RAxML, BEAST, DnaSP, Arlequin, Haplotype Network, TCS, and R (training in data manipulation and visualization).

- **Languages**

Kurdish (native) - Persian (native) - English (Fluent).

Work Experience

Environmental Research Scientist
Bina Azma Sepahan Company | Sep 2014 – Present | Full-time

- Conduct environmental sampling and laboratory analysis (air, water, soil, wastewater, food products).
- Perform data analysis and environmental risk assessment using statistical approaches.
- Contribute to pollution monitoring, wastewater treatment evaluation, and industrial environmental compliance strategies.

Certificate

Learn to Program Python: A 6-hour course delivered by Intersect Australia on Wednesday, August 31, and Thursday, September 1, 2022, at the University of Canberra**.**

Data Visualisation in R: A 3-hour course delivered by Intersect Australia on Friday, August 19, 2022 at Deakin University.

Data Manipulation in R: A 3-hour course delivered by Intersect Australia on Thursday, August 18, 2022 at Deakin University.

R Programming Software; Matris Pedia Training (MPT) Iran Representative, 2022.

Professional Activities

Attendance in t7th7st national conference on air pollution and sound management, Shahid Beheshti University, Tehran, Iran, (January. 28-29, 2019).

Member of the 7th national conference on air pollution and sound management, Shahid Beheshti University, Tehran, Iran, (January. 28-29, 2019).

English Proficiency (PTE Academic)

Overall: 63

Listening: 51 · Reading: 60 · Speaking: 71 · Writing: 55

References

Dr. Faraham Ahmadzadeh Associate Professor of Shahid Beheshti University, Tehran, Iran, Institute (ESRI).

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Dr. Yousef Rashidi Assistant Professor of Shahid Beheshti University, Tehran, Iran, Institute (ESRI).

Email: y_rashidi@sbu.ac.ir Phone: +98 21 29906019

Dr. Hanyeh Ghaffari Associate Professor of Kurdistan University, Kurdistan, Iran, Department of Environment, Faculty of Natural Resource.

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